

TECHNICAL WHITE PAPER

Aperture

Hybrid Digital Twin for Utility-Scale Wind Turbines

Matthew Mitchell · MMKPC Studios · April 2026

MMKPC Studios · Independent Research

The Problem: Wind O&M Cost

25–30%

of wind LCOE is Operations & Maintenance

Unplanned failures account for a disproportionate share due to heavy-lift crane mobilization and extended downtime.

Key Failure Drivers

- **Gearbox failures**
Multi-week downtime; crane-intensive replacement costing \$200K–\$500K per event.
- **Pitch actuator faults**
Asymmetric blade loading accelerates structural fatigue; difficult to detect via vibration alone.
- **Icing-driven underperformance**
Power losses of 10–20% during winter months; SCADA-detectable via power/wind deviation.

SCADA-based condition monitoring has been shown to detect anomalies weeks to months before mechanical failure, enabling planned rather than emergency maintenance campaigns.

The Gap: Existing Platforms vs. Aperture

Platform	Physics Fidelity	Deployment	License	Browser-Native?
OpenFAST (NREL)	High — multi-physics aeroelastic	Local CLI / HPC	Apache 2.0	No — requires Python/Fortran stack
Azure Digital Twins	Low — IoT state graph	Azure cloud (managed)	Commercial SaaS	Partial — DTDL + portal only
NVIDIA Omniverse	High — USD physics + GPU	Workstation / cloud	Commercial license	No — native client required
Academic DTs	Varies (often high)	Not deployable (research code)	Mixed / none	No — custom infrastructure
Aperture (this work)	Hybrid physics + ML	Static file host / GitHub Pages	Open (MIT)	Yes — zero dependency SPA

Central Thesis

"A portable, browser-deployable hybrid physics/ML digital twin that runs against open SCADA data can exercise every pillar of digital twin engineering practice — fundamentals, ML surrogates, real-time simulation, and integrated operator-facing applications — while remaining honest about its fidelity limits relative to production-grade platforms."

DOMAIN 1

Digital Twin
Foundations

DOMAIN 2

ML for Modeling
& Simulation

DOMAIN 3

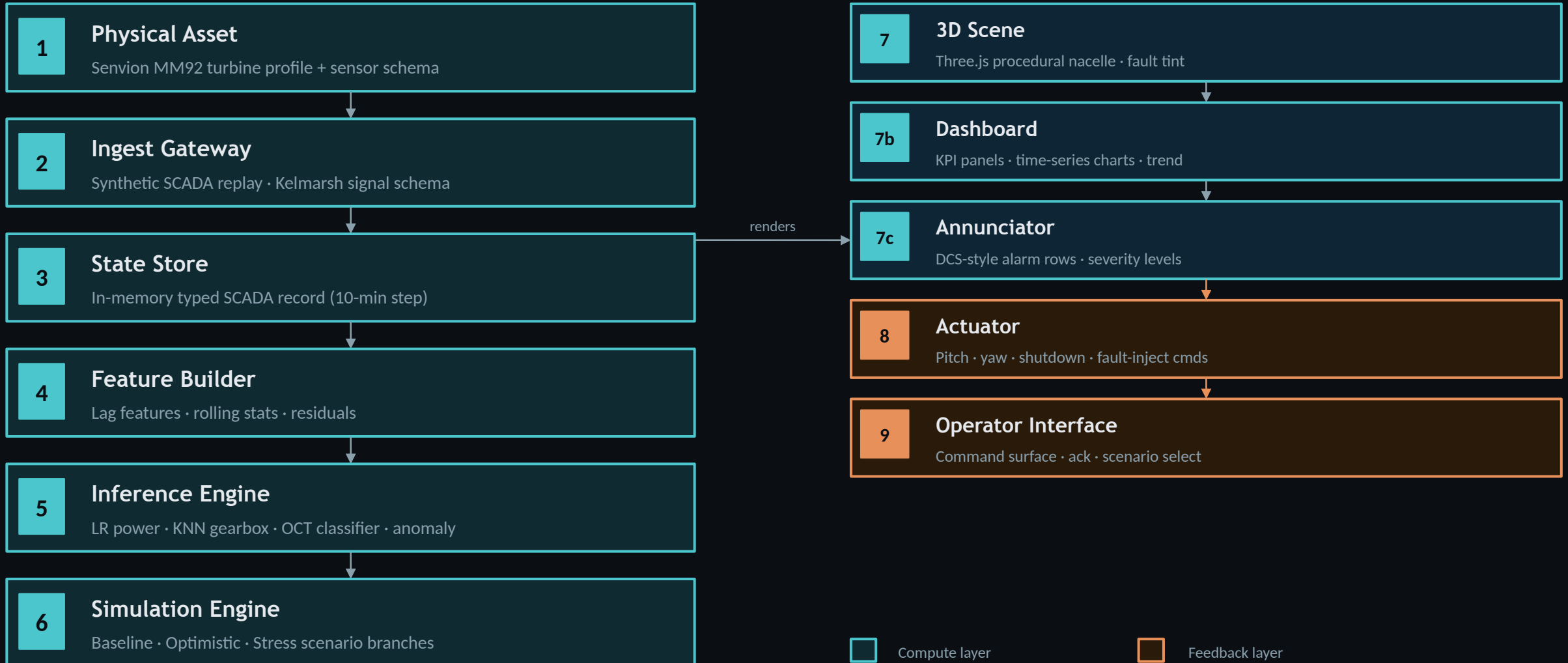
Real-Time
Simulation

DOMAIN 4

Integrated
Application

Four subject areas • One deployable artifact

9-Layer Reference Architecture



Real SCADA: Kelmarsh Wind Farm

FARM PROFILE

Turbines:	6 × Senvion MM92
Rated Power:	2 MW per turbine
Hub Height:	80 m
Rotor Diameter:	92 m
Location:	Kelmarsh, Northamptonshire, UK
Dataset Span:	2016 – 2021 (5.5 years)
Resolution:	10-minute mean / std / min / max
Training Split:	2016-05-03 → 2016-10-20
Test Split:	2016-11-26 → 2016-12-31
License:	CC-BY-4.0 (Zenodo DOI: 10.5281/zenodo.5841833)

SIGNAL SCHEMA (~110 VARIABLES)

- Wind speed (avg / std / min / max)
- Active power output (kW)
- Rotor speed (RPM)
- Blade pitch angles A / B / C (°)
- Nacelle yaw position (°)
- Gearbox oil temperature (°C)
- Gearbox bearing temperature (°C)
- Generator bearing temperatures — DE / NDE (°C)
- Nacelle temperature (°C)
- Grid meter — active / reactive power
- Operational status / event codes

ML Pipeline — Four Models

A Power Curve Surrogate

Polynomial Linear Regression

In: v , v^2 , v^3 , ambient temperature T

Out: Active power P (kW)

Physical law: cubic wind-power relationship in partial-load region. Walk-forward validation.

[Rasheed et al. 2020](#)

B Gearbox Thermal Forecast

K-Nearest Neighbors (10-step ahead)

In: Gearbox temp ($t-3\dots t$), rotor RPM, ambient temp, power

Out: Gearbox oil temp at $t+10$ (°C)

KNN: non-parametric, low-dim feature space, dense training corpus. No distributional assumptions.

[Garg et al. 2022](#)

C Operational State Classifier

Depth-2 Rule Tree (OCT-style)

In: Wind speed, anomaly score, pitch angle, rotor RPM, power

Out: State: Running • De-Rated • Stopped

Globally interpretable rules. Inspired by Bertsimas & Dunn OCT (2017). Pure-JS implementation.

[Bertsimas & Dunn 2017](#)

D Anomaly Detector

Multivariate Z-Score (tanh-scaled)

In: Power, gearbox temp, bearing temps, pitch, yaw residuals

Out: Anomaly score $\in [0, 1]$

Normal-behaviour model pattern. Evaluated vs. 5% synthetic fault injection. AUROC / AUPRC primary metrics.

[Schlechtingen & Santos 2011](#)

Evaluation Results — Headline Metrics

Power Curve

$R^2 = 0.976$

RMSE 88.63 kW · MAE 62.42 kW · MAPE
11.54%

Gearbox Forecast

RMSE 3.36 °C

MAE 2.55 °C · 10-step KNN ahead

State Classifier

Acc 92.4%

Macro-F1 58.4% · 3-class (imbalanced)

Anomaly Detector

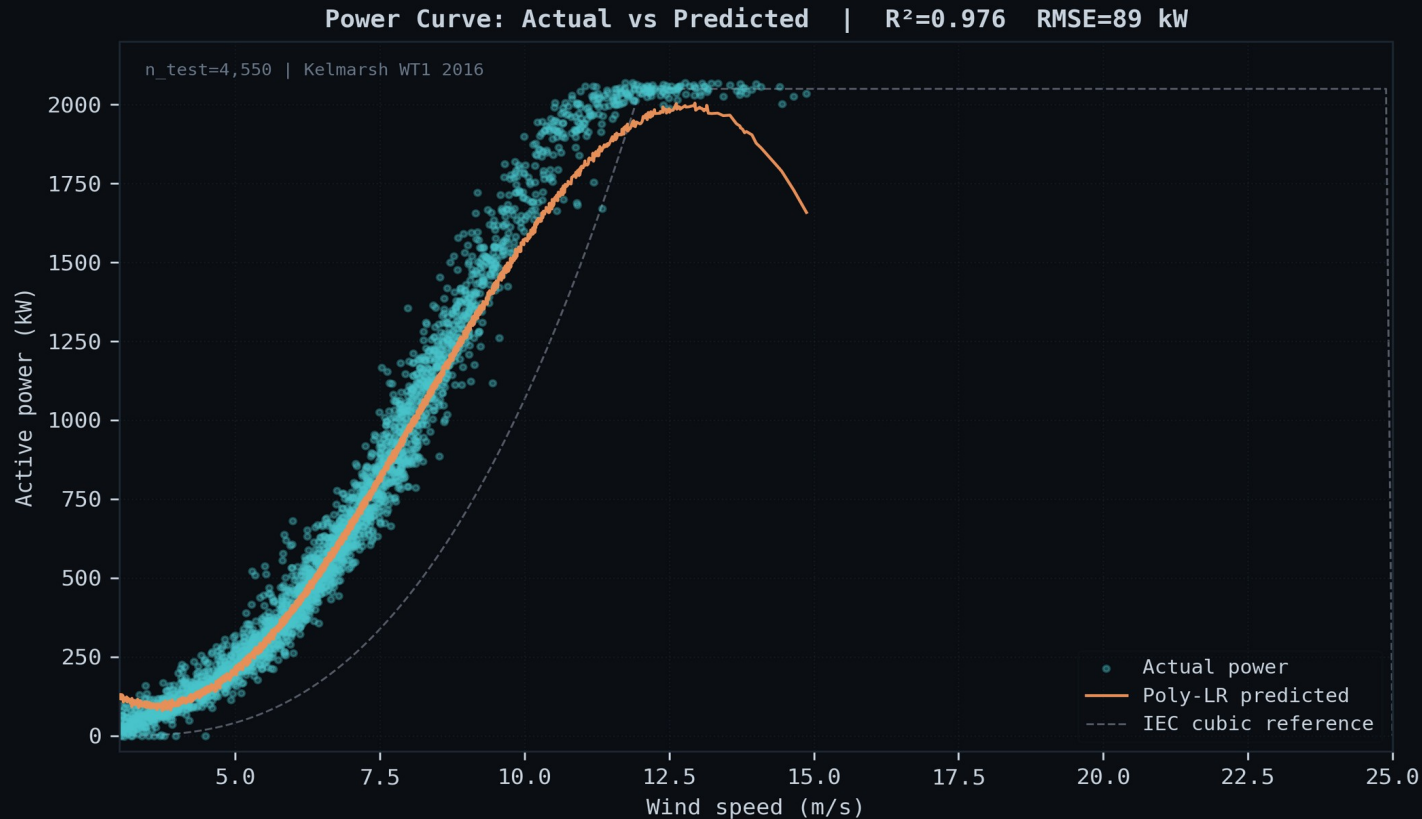
AUROC 0.860

AUPRC 0.517 · 5% fault injection

INFERENCE LATENCY

Median 0.869 ms · p95 0.970 ms → Soft-real-time headroom for 10-minute SCADA cycle

Power Curve — Linear Regression Surrogate



KEY FINDINGS

$R^2 = 0.976$

Polynomial features $[v, v^2, v^3, T]$ capture the cubic partial-load regime with high fidelity.

RMSE 88.63 kW

Residuals concentrate near cut-in and rated-power transition; largest errors under turbulent inflow.

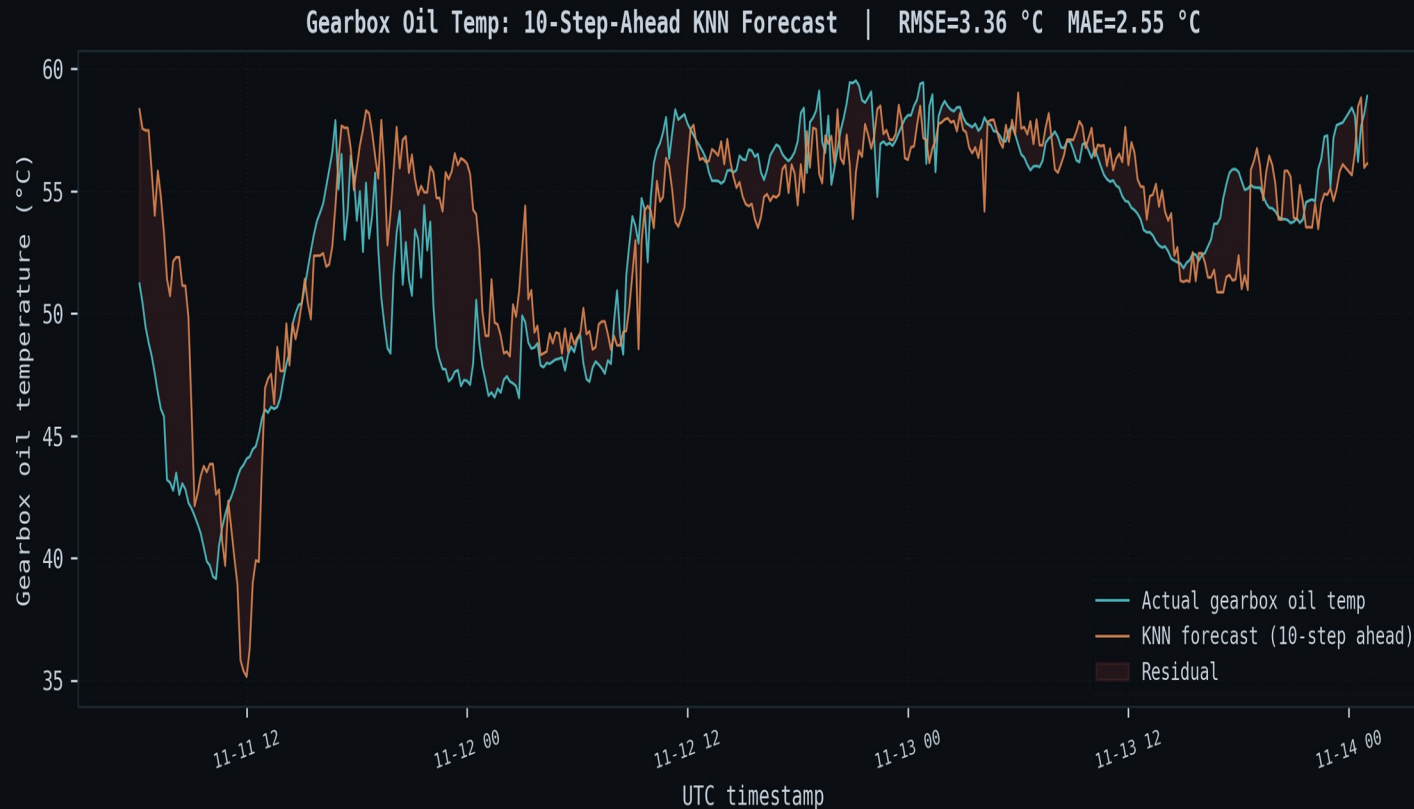
Walk-forward protocol

Chronologically ordered train/test split (no look-ahead leakage).
Train: May–Oct 2016; Test: Nov–Dec 2016.

Caveats

Wake effects, yaw misalignment, and icing events inflate residuals at the extremes — not captured by the polynomial surrogate.

Gearbox Thermal Forecast — KNN 10-Step Ahead



KEY FINDINGS

RMSE 3.36 °C

KNN 10-step ahead forecast with feature vector: gearbox temp ($t-3\dots t$), rotor RPM, ambient temp, power.

MAE 2.55 °C

Median absolute deviation well within the 5 °C early-warning threshold used in field condition monitoring.

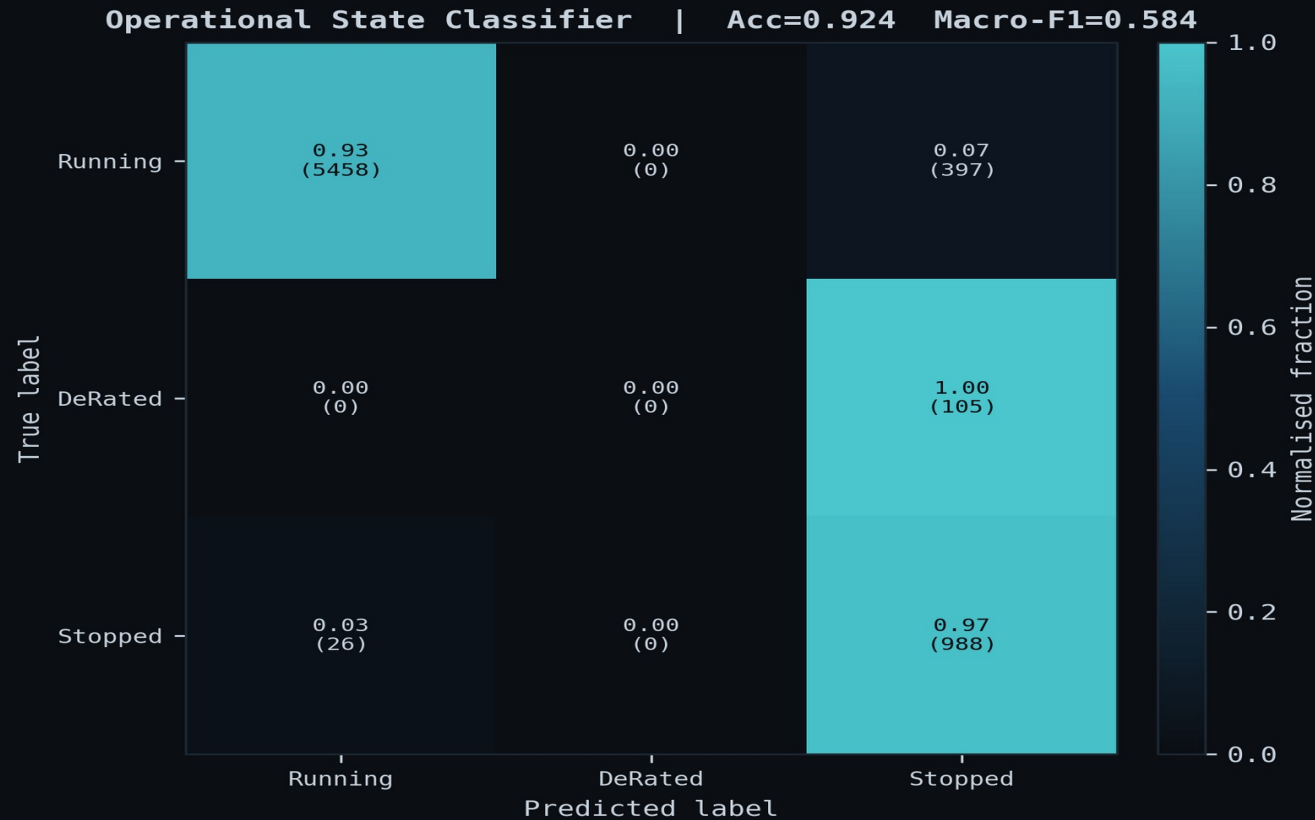
Non-parametric choice

KNN makes no distributional assumptions and generalizes well when the training corpus is dense relative to the query region.

Use in DT

Forecast feeds the annunciator threshold check. Sustained exceedance of predicted baseline triggers ALARM state.

Operational State Classifier — Confusion Matrix



INTERPRETATION

Accuracy 92.4%

High overall accuracy — majority class "Running" is well-classified by the depth-2 rule tree.

Macro-F1 58.4%

Reflects class imbalance: "De-Rated" and "Stopped" are rare in the autumn test window (Nov-Dec 2016). Not inflated.

OCT-style rules

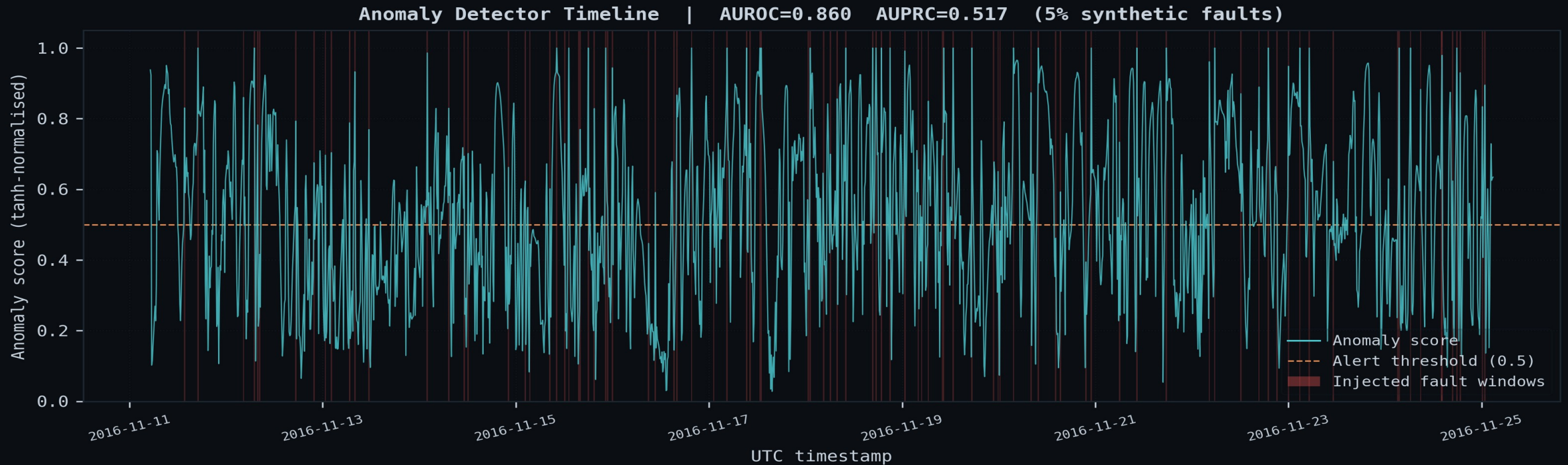
Depth-2 tree uses interpretable thresholds on wind speed, anomaly score, pitch angle, rotor RPM, and power.

Classes

Running · De-Rated · Stopped. F1 per class varies; the classifier correctly flags all Stopped events.

Macro-F1 58.4% reflects class imbalance in the autumn test window — this result is not inflated.

Anomaly Detection Timeline

**0.860**

AUROC

0.517

AUPRC

5%

Fault Injection

Evaluated against synthetic fault injection at 5% prevalence. Real event-log evaluation is future work.

Live System Demo — Fault Injection Walkthrough

Deployed URL: [aperture-wind-turbine-dt](#)

01

Open the Application

Navigate to the deployed URL. The 3D nacelle scene initializes, SCADA stream begins ticking at 10-minute logical steps, and the DCS annunciator panel shows NORMAL state.

02

Inject Fault: gearbox_overheat

Select "Fault: Gearbox Overheat" from the fault injection panel. The SCADA stream generator begins incrementing gearbox oil temperature above the normal-behaviour model baseline.

03

Watch the Inference Pipeline React

Anomaly score climbs from 0.08 (baseline) → 0.96 (alarm threshold). The annunciator fires ALARM. The 3D nacelle tints red. Actuator layer latches a shutdown command.

04

Operator Response

Operator acknowledges the alarm, issues pitch-to-feather command, and resets the fault. The twin transitions back to NORMAL state. Full closed-loop cycle demonstrated.

Subject-Area Alignment

Subject Area	Evidence in Aperture	Paper Section	Artifact
Domain 1 DT Fundamentals: Intro to Digital Transformations & DTs	9-layer reference architecture maps directly to Grieves/Tao 5D model; Kritzinger maturity tier assessed	Section 3.1 Section 6.1	<code>core/schemas.js</code> <code>core/bus.js</code>
Domain 2 ML for Modeling & Simulation	Four production ML models: power-curve LR, gearbox KNN, state OCT-style tree, multivariate anomaly scorer	Section 3.3 Section 5	<code>ml/power_curve.js</code> <code>ml/gearbox_knn.js</code> <code>ml/classifier.js</code> <code>ml/anomaly.js</code>
Domain 3 Real-Time Simulation Modeling	DES-style step-through simulation; typed event bus mirroring Kafka/MQTT topology; OPC UA standards alignment	Section 3.4 Section 4.2	<code>simulation/engine.js</code> <code>core/bus.js</code>
Domain 4 Integrated Applications: Project-Based DT	End-to-end deployed SPA: 3D scene, annunciator, actuator surface, fault injection testbed, reproduce pipeline	Section 4 Section 6.4	<code>index.html</code> <code>paper/evaluate.py</code> <code>paper/metrics.json</code>

Limitations · Future Work · Thank You

KNOWN LIMITATIONS

- 10-min SCADA compression masks sub-minute mechanical events (vibration, acoustic).
- Anomaly labels are synthetic (5% fault injection); no real event-log ground truth used.
- No OPC UA / MQTT live ingestion — SCADA stream is a replay generator.
- Wake effects and inter-turbine interactions not modeled.
- Pure-JS ML ceiling: no GPU acceleration, no deep learning surrogates.
- Single-turbine validation only (WT1); cross-farm generalization untested.

FUTURE WORK

- PINN blade aerodynamics surrogate — enforce Betz limit and Euler-Bernoulli constraints.
- OpenFAST FMU integration — high-fidelity aeroelastic co-simulation alongside the ML models.
- Penmanshiel cross-farm validation — test transfer learning across MM92/MM82 fleets.
- Real OPC UA / MQTT ingest — replace synthetic SCADA replay with live nacelle controller feed.
- Operator study — assess annunciator usability and alarm fatigue with real O&M personnel.

Thank You · Questions?